

Good morning everyone. Today I will be presenting our analysis of risk factors for stroke using data from the Framingham Heart Study. This is a classic dataset in cardiovascular epidemiology.

So the investigator asked us to address two questions. The primary goal was to identify which risk factors predict 10-year stroke and then use those models to compute predicted probabilities for clinically meaningful risk profiles — for example, what is the 10-year stroke risk for a 60-year-old diabetic man with high blood pressure versus an average-risk woman? We did this separately for men and women. The secondary aim was descriptive: the data include three examination periods spanning about 12 years, so we wanted to see how much risk factors actually change within the same individuals over time. Now, regarding covariates: three variables were forced into every model based on established clinical evidence — age, diabetes, and systolic blood pressure. The five candidate variables we were testing are prevalent CHD, BP medication use, current smoking, total cholesterol, and BMI. Our null hypothesis is that after adjusting for the three variables, none of these five candidates add predictive value for stroke.

Our data come from the Framingham Heart Study, which enrolled over 5,000 participants starting in 1948. The subset we received includes 4,434 participants across three examination periods spanning roughly 1956 to 1968. We restricted our analytic cohort to the 4,402 participants who were stroke-free at baseline. As you can see in Table 1 on the right, the cohort is split into 1,930 men and 2,472 women, with a mean age of about 50 in both groups. A few key differences between the sexes: men had much higher smoking prevalence at 60.5% versus 40.5% for women, and more prevalent CHD at 6.2% versus 2.7%. Women had slightly higher systolic blood pressure. Now, an important feature of this data: within 10 years, only about 2.5% in each sex experienced a stroke — so this is a relatively rare event with only 111 total cases. But, 208 men — over 10% — and 159 women — about 6.4% — died without having a stroke first. This death-without-stroke is a competing risk. Our Cox models treat these deaths as non-informative censoring, which could slightly overestimate stroke incidence. Missing data was minimal — BPMEDS had the most at about 1 to 2 percent — so we used complete-case analysis.

For our statistical approach, we used Cox proportional hazards models fitted separately for men and women. The outcome is time to first stroke, censored at 10 years. As I mentioned, three variables — age, diabetes, and systolic BP — were forced into every model. The five candidate variables were then evaluated for inclusion. For model selection, we used AIC-based stepwise selection, running it in both backward and forward directions. Running both directions lets us check whether they converge on the same model, which would give us confidence that the selection is robust. For inference on the final model, we used the Wald test at alpha equals 0.05. We verified the proportional hazards assumption using Schoenfeld residuals, and assessed model discrimination with the concordance index. For the time-varying analysis, the investigator wanted to know how much risk factors change over the three examination periods. A naive approach would just compute means at each period, but the composition of participants changes across periods due to dropout and death. So we restricted to a balanced cohort — only participants present in all three periods — which removes that confounding. We then computed within-individual paired changes, Period 3 minus Period 1, to directly measure person-level change rather than just comparing cross-sectional means.

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Now for the results. First, a key finding on model selection: backward AIC, forward AIC, converged on exactly the same variable sets for both sexes. This is strong evidence that the selection is robust. For men, the final model retained four variables. The strongest predictor was diabetes, with a hazard ratio of 4.807, meaning diabetic men had nearly five times the stroke hazard compared to non-diabetic men. Age and systolic BP were also highly significant. Current smoking had a hazard ratio of 1.924 — so about double the risk — and was the only candidate variable that improved the model. One interesting finding: the diabetes hazard ratio jumped from 3.535 in the base model to 4.807 after adjusting for smoking. This tells us smoking was partially confounding the diabetes-stroke association, attenuating it in the base model. For women, no candidate variable improved AIC. Diabetes had a hazard ratio of 1.767 but was not significant at $p = 0.275$ — this is likely a power issue since very few diabetic women experienced stroke events, not a true null association. The proportional hazards assumption was satisfied for both models.

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These two figures illustrate the results visually. Our secondary analysis of within-individual risk factor changes. We restricted to a balanced cohort of 3,139 participants present in all three examination periods, so we are not confounding time trends with changes in who is observed. The results: systolic BP rose about 10 mmHg on average, diabetes prevalence roughly quadrupled, over 20% of male smokers quit by Period 3, and BP medication use increased dramatically. These substantial changes mean that baseline values measured at Period 1 become increasingly misaligned with what participants are actually experiencing over the 10-year follow-up, which supports that a time-varying covariate analysis would be worth pursuing.

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To wrap up. Our analysis identified age, diabetes, and systolic blood pressure as the primary determinants of 10-year stroke risk in both sexes. Current smoking added independent risk in men with a hazard ratio of about 1.9, but was not retained for women. The within-individual changes we observed strongly support the investigator's suggestion that a time-varying covariate analysis could yield different and potentially more accurate results. For limitations: first, deaths without stroke are treated as non-informative censoring in standard Cox models, and given that over 10% of men died, this could overestimate stroke incidence. Second, our models rely on baseline-only covariates that, as we showed, become increasingly misaligned with true exposure. Finally, the Framingham cohort is predominantly white, so generalizability to more diverse populations may be limited. Thank you very much. I am happy to take any questions.